
Networking and Cloud Fundamentals

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1 INTRODUCTION

In this assignment, you will work with networking and remote access tools using both the GUI and CLI. You will check your IP address with and without a VPN, create SSH keys, and log in to the LIACS server using key-based authentication. You'll also learn about setting up a GitHub repository.

You need to hand in your answers on Brightspace in a ZIP file. Make sure all filenames in the ZIP archive are clear and descriptive (including screenshots). Check the schedule for the deadline.

2 OPERATING SYSTEM

This assignment can be done on Windows or Linux-based (Ubuntu). When you see the following icons appear, read the instructions for your operating system.

 **Windows:** Windows instructions.

 **Linux:** Linux instructions, they might also work on MacOS.

3 INITIAL SETUP

Before you start, create the file "answers.txt" in your lecture 5 folder. Here, you will write down all the answers to the questions in this assignment. You will need to include this file in the final zip that you will upload to Brightspace.

4 IP ADDRESS, ROUTING & WEBREQUESTS

In this task, you will look at IP addresses using various commands in a CLI.

1. Write down your IP address. Do this by making an internet request, using `curl` to the following site `ifconfig.me`. Also, check your IP address locally and compare the two. This can be done using the command `ipconfig` or `ip route`. Here, you can find also information of your default gateway, write this down as well.
2. Repeat the previous step, but this time with your VPN turned on. Do you notice any changes in the result? If so, explain why it changed — or why it didn't.
3. Is your IP address private or public, and what about your gateway? Private IP addresses are in the range 10.0.0.0 - 10.255.255.255, 172.16.0.0 – 172.31.255.255, or 192.168.0.0 – 192.168.255.255.
4. Find out the geo-location of your IP address. You can use the following webrequest `curl ipinfo.io` or the location of any other IP address `curl ipinfo.io/<IP Address>`.
5. Trace the internet hubs your data passes through to reach a website like Google. Use the command `tracert google.nl` or `traceroute google.nl` to see the path it takes. What do you notice about the first hub and the last hub?
6. Find the geo-location of the Google data center that is used for your Google searches?

5 CREATE GIT REPO

In this task, we will create an SSH key and use it to create a connection to a git server. For this task, you can either use [LIACS Gitlab](#) or [Github](#); the process will be the same. An SSH key has two parts, a private part known only to you and a public part that you use to identify yourself. The public part can be installed at servers and when you login, the server encrypts a message with the public key. Your computer then decrypts it with your private key, and the result is sent back so the server can verify it is you. No more need for passwords.

5.1 SSH KEY CONNECTION

The first task is to ensure that you have an SSH key on your system. So, first check if you have an ed25519 SSH key, see [check for ssh key on gitlab](#) or [check for ssh key on github](#) and follow the steps. If this is not the case, continue by making an ed25519 SSH key, see [generate ssh key on gitlab](#) or [generate ssh key on github](#) and follow the steps. *To access these links you need to login at [LIACS Git](#) with your ULCN account first*

Next, we need to add the public part of the SSH key to your preferred git server. Go to user settings SSH keys on your preferred git server; [Gitlab](#) or [Github](#). Click add new key or new SSH key. Now, add the public part of your SSH key in the key field and give the key a title (so you can remember which system the SSH key is from). The public part of the SSH key can be found in the .ssh folder in the file "id_ed25519.pub". Open the file (with a plain text editor) and copy the contents to the key field.

5.2 CREATE REPOSITORY

At this point, we can communicate with your preferred git server using SSH. However, we need to set up a git repository to be able to send and receive files for a certain project. In this part of the task, you will essentially make a remote backup of your local course directory.

Go to the git server website (homepage) and click "new" or "+" icon to create a new blank git repository. Give the repository a logical name like "DSIP" and optionally give it a description. Click "create repository" to finish the process.

Now, we can clone the project to our local machine. Under the button "Code", you can copy the SSH link to your project. Now, clone the project in the same folder as your "DSPI" root folder. You can do this by going to that folder in your terminal and type `git clone <GIT LINK>`.

For git to work properly, it is necessary that the git server knows who is adding data to the server. Therefore, you need to store in the configuration files who you are. This can be achieved using the following two commands (without it, git commit will not work):

- `git config --global user.name <YOUR NAME>`
- `git config --global user.email <YOUR@EXAMPLE.COM>`

Next, copy all content from your original "DSIP" folder to the new git folder. Remember you can do this with `move <ORG_DSIP_FOLDER>\* <GIT_FOLDER>\`

`mv <ORG_DSIP_FOLDER>/\* <GIT_FOLDER>/ .`

Do not continue, before making sure that everything is moved!!!. Now, you can remove the old folder if it is empty.

Locally, the Git folder contains new files and folders, but if you check the Git server (e.g., the website), you can see that those files and folders aren't there yet. To make Git aware of the changes in the local folder, use `git add .` (note: this adds all changes). Then, create a new commit to take a snapshot of the current state, `git commit -m "<MESSAGE>"`. Choose an appropriate message. Finally, push the changes to the server so they appear remotely, `git push`.

You can also retrieve changes from the Git server to your local machine. To see how this works, start by adding a new file to the Git server named server.txt. You can do this by clicking the "+" icon or selecting "Add file" at the git website. Name the file server.txt and enter the content: hello git. Then click "Commit changes" and write a commit message, such as "added server.txt".

At this point, the file exists on the Git server, but it hasn't been added to your local repository yet. To update your local copy, use the command `git pull`. This will fetch the latest changes from the server, and you should now see the server.txt file in your local directory.

5.2.1 GIT SHELL SCRIPT

In lecture 4, we learned that we can group CLI commands in a shell script. Uploading something to the git server is a typical example, where each time you want to upload something to the server you use the same three commands:

1. `git add *`
2. `git commit -m "<MESSAGE>"`
3. `git push`

So, make a shell script (`bat .bat` `bash .sh`) that contains these three commands and then run the script to upload this new file to the git server.

5.2.2 GIT DELIVERABLES

Make two screenshots of your project on the git server. One should show the main folder with all its subfolders, and the other should show the files in your "lecture 5" folder.

6 LIACS SERVER

In this task, you are going to log in to the LIACS server and put some files on the server.

Important: You need to be either on the university wifi (eduroam) or enable the VPN to access the LIACS server, "silver".

6.1 LOGIN

First, we are going to make a connection with the server using the SSH protocol. To do this open your terminal and type `ssh <ULCN>@ssh.liacs.nl`. It will ask to trust the server, type `yes`. Now, enter your ULCN password to connect. This should give you access to the silver server.

Remarks:

- *Copy pasting in a Linux terminal (the server is Linux, Apache) is CTRL+SHIFT+C/V.*
- *Please contact us, if you can not log in because your username is not recognized. This can happen if you do a double bachelor's or are enrolled in another bachelor's program for any other reason.*

6.2 SSH SETUP

We can again use your SSH keys, to prevent typing your password every time you want to login or transfer file from/to the server. To do this create a folder on the server (in your directory) and name it ".ssh". Now, go to this folder and type `nano authorized_keys`. This will open a plain text editor in the terminal (running on the server) with the filename "authorized_keys".

Now, copy your public SSH key (same as for git) in this file and save & close the file (`CTRL+X` , select "y"). You can check if the content of the file is correct by typing `cat authorized_keys` ; this should display your SSH key.

6.3 TRANSFER FILES

Exit the server (`exit`) and go to your lecture 5 directory. Now, copy your "answers.txt" to your personal folder on the LIACS server. You can copy files using the following command

`scp -r <FILEPATH> <ULCN>@ssh.liacs.nl:/home/<ULCN>/` . Note, that "FILEPATH" can be absolute or relative and the "-r" flag is only needed to copy folders.

Make a screenshot of the content of your "answers.txt" on the LIACS server. Remember, you can show the content of a file in the command line by typing `cat <FILENAME>`